

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Specifications for Concrete Weight Coating
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Specifications for Concrete Weight Coating

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Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	06-06-22	First Issue for Clients Review
R02		
A01		

Review Code & Status (to be filled by Client)		
☐ Code 1	Approved – Further work may proceed.	
☐ Code 2	Approved with comments. – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed	
☐ Code 3	Reviewed – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed	
☐ Code 4	Retained for information only	
☐ Code 5	Review Not Required – Do not Resubmit	
Date:	Position:	Sign:

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- PPL-SPC-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Wall Thickness Calculation Report
EA-BFPCL-GPMC-GGPL-FEED-PPL-ALS-001 to 036	Pipeline Alignment Sheet
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-003	Pipeline Buoyancy / On Bottom Stability Analysis
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-009.	Specification for Pipeline Construction.

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Abbreviations

ABBREVIATIONS	MEANING
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
CS	Carbon Steel
DEP	Design and Engineering Practices
DP	Design Pressure
DWT	Drop Weight tear
FEED	Front End Engineering
HAZ	Heat affected zone
ISO	International Organisation for Standardization
LTO	License To Operate
3LPE	Three-Layer Polyethylene
MMscf/d	Million standard cubic feet per day
NDE	Non-Destructive Examination
NPS	Nominal Pipe Size
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
TD	Design Temperature
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

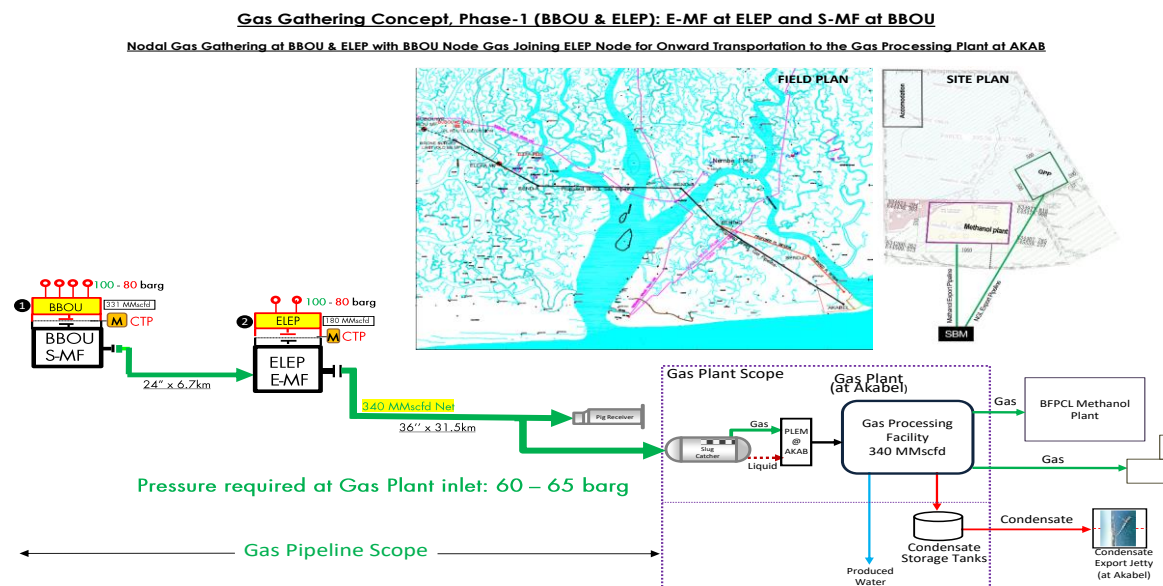


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The Shell DEP. "31.40.30.30-Gen (Oct. 1999)- Technical Specification-Concrete Coating of Line pipe" shall be the primary document to be used for the design requirement, and the application of the concrete coating on the line pipes to be used at water course crossing, swamp and marsh areas for BFPCL Gas gathering Pipeline project-phase 1.

This Specification elucidates amendments and supplement to the Shell DEP. 31.40.30.30-Gen.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP. 31.40.20.37-Gen	Line pipe for Critical Service (amendments/supplements to IOGP S-616 AND DNVGL-ST-F101 2019)
DEP. 31.40.30.30-Gen	Technical Specification-Concrete Coating of Linepipe.
DEP. 31.40.40.38-Gen	Hydrostatic Pressure Testing of new Pipeline (February 2017)
DEP. 61.40.20.30-Gen	Welding of Pipelines & Related Facilities (Amendments / supplements to API Std 1104)
ASME B31.8	Gas Transmission and Distribution Piping Systems
API 5L	Specification for Line Pipe
API RP 5LW	Recommended Practice for Transportation of Line pipe on Barges and Marine Vessels
ISO 3183	Petroleum and Natural Gas Industries – Steel Pipe for Pipeline Transportation Systems

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs

- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Line Pipe Data

3.1. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Table 3.1: 24" Multiphase Pipeline Data

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength (SMYS)	483 Mpa
NPS	24"
Density	7,850 kg/m ³
Young's Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570 MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
OD / Wall Thickness	610/15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	60mm
Installation Condition	Trenching
Cathodic Protection	Impressed current

3.2. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

Table 3.2: 36" Multiphase Pipeline Data

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	36"
Density	7,850 kg/m ³
Young's Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
OD / Wall Thickness	914/23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	60mm
Installation Condition	Trenching
Cathodic Protection	Impressed current

4. Environmental and Pipeline Design Data

Refer to the document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 - Basis of Design

4.1. Location Requiring Pipe Concrete Weight Coating

4.1.1. Water course Crossings

At watercourse crossings of 6.0 metres or less in width, no concrete weight coating is required if the remaining adjacent portion of the ROW is dry land. At watercourse crossings, greater than 6.0 metres in width, concrete weight coating shall be applied for the high water level width plus a minimum of 5.0 metres extra width on each side of the high water level. High water level shall be based on 25 years environmental return conditions. Contractor shall design the overweighing of the pipeline installed at major navigable rivers; the pipeline stability at these crossings shall also be verified against the drag forces on the basis of stream current velocity.

The concrete weight coating applied at water course crossings shall be of sufficient thickness to ensure a minimum negative buoyancy factor of 1.25 based on the established density of the water. At water courses where the rate of flow may exceed 2.5 m/s during the rainy season and/or the soil at the bed of the watercourse can be reasonably described as more than 50% sand (by particle size analysis) the buoyancy factor (K) shall be increased to a minimum of 1.5.

4.1.2. Swamp and Marsh Areas

At crossing of swamp and marsh areas, the pipeline shall be continuously concrete coated as for the watercourses to ensure a minimum negative buoyancy factor of 1.25.

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5. Concrete Coating Design and Application

Concrete coating of line pipe shall be applied in accordance with the requirements of DEP 31.40.30.30-GEN and the amendments contained in section 6 of this document. These shall also to be reflected on the alignment sheets.

5.1. Density of Water

For the buoyancy calculations, density of water at marshy/swampy terrains shall be 1150kg/m³, the standard density of sea/brackish water.

6. Amendment to DEP 31.40.30.30-GEN

Hereinafter the modification made to DEP 31.40.30.30-GEN (Dec 1999) to be applied in the project.

6.1. DEP Section 1 - Introduction

Sub Section 1.1 Scope

The concrete class shall be Grade B for installation in estuaries, rivers, creeks and swamp areas.

Sub Section 1.3 Definitions

6.2. DEP Section 2 – Materials

Sub Section 2.6.1 General.

Delete from the 2nd paragraph the following:

- The Principal shall specify, in the Scope of Work, the type of reinforcement to be used.

Added at the end of 2nd paragraph the following:

- Contractor shall propose to Company, for approval, the selected type of reinforcement.

Sub Section 2.6.3 Cage Type

Delete from the 5th paragraph the following:

- Cages shall have two hoops spaced 50 mm apart at each pipe end and adjacent to anodes and arrestors.

Added at the end of 5th paragraph the following:

- Cages shall have two hoops spaced 50 mm apart at each pipe end. Delete the NOTE at the end of 5th paragraph

6.3. DEP Section 4 - Coating Process

Sub Section 4.1 Coating Procedure Specification

Delete from the list of coating procedure the following:

- anodes installation
- crack/buckle arrestor installation.

Sub Section 4.2 Coating Procedure Qualification Testing

Delete from the list of activities the following:

- Installation of at least 1 anode including 3 Thermit welds

- Coating of 1 pipe with an anode.

Sub Section 4.4.1 Pre-checks

Delete the 2nd paragraph the following:

- The testing shall take place just prior to the fitting of the reinforcement steel and prior to the attachment of anodes or buckle/crack arrestors. Any pipes containing pipe or coating defects shall be put aside or repaired in accordance with approved repair procedures before the fitting of the reinforcement steel anodes or arrestors.

Replace the 2nd paragraph with the following:

- The testing shall take place just prior to the fitting of the reinforcement steel. Any pipes containing pipe or coating defects shall be put aside or repaired in accordance with approved repair procedures.

Delete the 3rd paragraph

Sub Section 4.4.3.1 General. Delete the 3rd paragraph

Paragraph 4.4.3.5 End cut back and tolerance.

Delete from the paragraph the following:

- The length of this "cut back" i.e. uncoated section shall be as specified in the Scope of Work with a tolerance of +/- 15 mm. The concrete shall be terminated either square or tapered back at an angle to the pipe surface as specified in the Scope of Work.

Added to the paragraph the following:

- The length of this "cut back" i.e. uncoated section shall be of 300 mm with a tolerance of +/-15 mm or at least a minimum of 150mm cut from the 3LPE coating. The concrete shall be terminated tapered back at an angle of two-to-one slope to the pipe surface.

Sub Section 4.5.4 Curing by sealing compounds (Membrane).

Delete 3rd paragraph

6.4. DEP Section 5 - Inspection and Testing

Sub Section 5.1 General

Add at the end of the Sub Section the following:

- Company shall have the right, at any time to inspect, test and examine the materials and equipment used in the manufacturing process. Any material or equipment, which is found not in compliance with the requirements of this specification, shall be replaced to the satisfaction of Company. The costs of these replacements shall be solely borne by Contractor.

- Company shall have free access to all Contractors' coating operations and the Contractor's sites for storage of materials and pipe.
- Company shall have the right to reject all materials, equipment and work which do not conform to these specifications.
- Concrete coated pipes not meeting the requirements as outlines herein, shall be rejected, stripped and re-coated by Contractor.

Sub Section 5.4 Determination of Submerged:

Weight 1st formula:

- Effect of 3 mm of extruded polyethylene corrosion coating with a cut back of 150mm +20/- 0mm at each end shall be incorporated in the equation as non-variable value.

-

Sub Section 5.6 Impact Testing:

This sub section is not applicable.

6.5. DEP Section 7 - Installation of Anodes

This section is not applicable.

6.6. DEP Section 8 – Pipe Identification

Add at the end of the subsection the following:

Under 8.3 Pipe Tally

- A certified permanent record / tally shall be supplied to Company showing the mill data and all weights, date of concrete coating, test and inspection done with results, potential repairs and special applications using the pipe mill serial number sequence in combination with the sequence of the concrete coating process, etc.
- The certified permanent records / tally shall be submitted to Company both on hard and soft copy (e.g.: on Excel file version).

6.7. DEP Section 9 - Documentation And Approvals

Delete from the 5th paragraph the following sentences:

- For pipe fitted with an anode, the anode shall not be in contact with any pipe, concrete or object that could damage the anode or its connections.

6.8. DEP Section 10 - Documentation

Sub Section 10.2 Records

Add to the end of first paragraph the following:

- Contractor shall be required to operate a pipe tracking procedure.

Sub Section 10.3 Logging and recording of data

Delete point A, B, C and D from the list of data and substitute as follows:

A. Line pipe:

- Serial number
- Nominal Pipe Size
- Pipe length
- Pipe weight
- Wall Thickness
- Date of entry into Contractor's Concrete Coating Yard
- Note of condition of received pipe

B. Anticorrosion coating

- Holiday detection method of 3 Layer polyethylene coating
- Repair procedure for polyethylene coating and concrete coating

C. Concrete Coating

- All measurements used in weight and volume calculation
- Calculated Volume
- Concrete details (incl. aggregate used, mixture ratio and density)
- Target Concrete Thickness
- Crushing strength test data if measured
- Agreed values, quantity and percentage of weight, volume and BSG of concrete coating components.
- Place and Date of Concrete Coating and acceptance date
- Curing method and time
- Actual Weight, if weighed
- Date of leaving Contractor's Coating Yard
- Notes on condition on leaving Contractor's Coating Yard
- Colour Code
- And any further data that may be required by the Principal."

6.9. DEP Section 11 – Summary of Required Principal's Approvals

Delete from the list of documents the Clause 7 relevant to Installation of anodes.

6.10. DEP Section 12 – Summary of Inspections/Tests

Delete from the list of documents the Clause 7 relevant to Installation of anodes

6.11. DEP Section 13 – Scope of Work Checklist

Delete from the list of documents the Section/Clause relevant to Installation of anodes.

Section reference	Description	value
1.1	Concrete Grade	B
2.1.1	Pipe specification - Pipe data: - Pipe diameter (external diameter constant) - Wall thickness (24"/36") - Steel grade - Pipe tally details (length, weight, heat no)	EA-BFPCL-GPMC-GGPL-FEED- PPL-SPC-001 24":610 mm, 36": 914mm 15.88/ 23.83mm API 5L Gr-X70 Note (1)
2.1.3	Internal pipe cleaning prior to load out. Internal clean less grade (ISO 8501-1)	Not required Not required
2.6.1	Reinforcement type, see note (2)	Welded wire/re-bar cage
4.2.1	Anti-corrosion coating specification - Anti-corrosion coating thickness - End cut-back of anti-corrosion coating	Not applicable 3mm minimum 150 mm (+20/-0mm)
4.4.3.5	Concrete termination End cut-back of concrete	Tapered at two-to-one slope 300 mm (+/-15mm) min.
5.2.2	Concrete thickness, see note (3)	(+/- 5mm)
5.4	Submerged weight requirement (unsaturated-as applied) - nominal submerged weight, - individual pipe tolerance - 25 pipe analysis tolerance - shift production average tolerance - Allowable variation per pipe joint (weight in air)	see note (4) 150 N/m +/- 4% +/- 2% Not applicable
5.6.1	Impact testing	Not applicable
7	Anode design drawing number	Not applicable
7.3	Anode bonding leads supplied loses	Not applicable

8.3	Colour code,	See note (5)
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Note (1): To be specified by Contractor QA/QC team. Note (2): Poultry net or chicken wire shall not be used

Note (3) & (4): Contractor shall refer to the Project basis of design EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 referenced in section 2.2

Note (5): Concrete Coater Contractor shall define and notify of the colour code.